CUSTOMER CHURN PREDICTION

Machine Learning Project Report

EDA | Decision Tree | Random Forest | XGBoost | Cross-Validation | Hyperparameter Tuning

Project Goal

Predict telecom customer churn, compare multiple classification models, and translate model outputs into actionable retention insights.

Dataset: IBM Telco Customer Churn | 7,043 customers

Final tuned XGBoost test performance: 79.9% accuracy | 65.9% precision | 50.5% recall | 57.2% F1

Executive Summary

This project develops an end-to-end machine learning workflow for predicting customer churn in a telecom setting. The analysis begins with data quality checks and exploratory data analysis, continues through preprocessing and model development, and ends with hyperparameter tuning, final holdout evaluation, feature importance analysis, and business recommendations.

The dataset contains 7,043 customers and a moderately imbalanced target: 26.5% of customers churned. Several high-risk patterns were identified during EDA, especially month-to-month contracts, electronic-check payments, fiber-optic service, lack of technical support, and lack of online security.

Three supervised models were compared with 5-fold cross-validation. Random Forest achieved the highest cross-validation accuracy and precision, while XGBoost achieved the highest recall and F1-score. XGBoost was therefore selected for tuning with RandomizedSearchCV. The tuned model achieved a best cross-validation F1-score of 0.595 and a final test F1-score of 0.572.

Key Takeaway

The final model provides useful predictive signal, but recall remains near 50%. In a real retention program, threshold tuning and cost-sensitive evaluation would be important before deployment.

Report Structure

- 1. Introduction: Business Problem and Dataset
- 2. Data Quality and Exploratory Analysis
- 3. Preprocessing and Modeling Strategy
- 4. Model Comparison and Tuning
- 5. Final Evaluation
- 6. Feature Importance and Business Insights
- 7. Limitations, Future Work, and Conclusion

1. Introduction: Business Problem and Dataset

1.1 Business Problem

Customer churn occurs when a customer stops using a company's services. For subscription businesses, identifying customers at high risk of churn can support targeted retention campaigns and help reduce lost revenue.

The modeling objective is binary classification: predict whether each customer will churn (1) or remain with the company (0).

1.2 Dataset Overview

Item	Value
Observations	7,043 customers
Original variables	21 columns
Target	Churn (Yes/No)
No churn	5,174 (73.5%)
Churn	1,869 (26.5%)
Problem type	Binary classification

The original variables describe demographics, customer tenure, telecom services, support services, contract type, billing behavior, and charges.

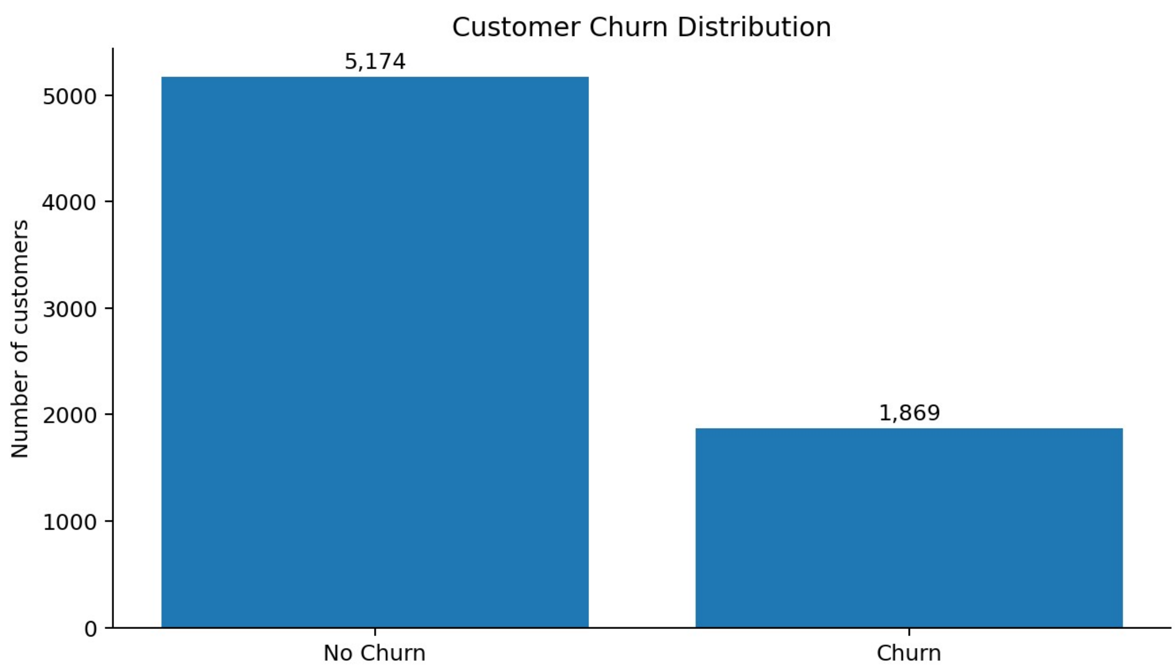

Figure 1. Target distribution showing moderate class imbalance.

2. Data Quality and Exploratory Data Analysis

2.1 Data Quality Checks

- Standard null checks initially reported no missing values.
- A systematic scan of text columns identified 11 whitespace-only values in TotalCharges.
- All 11 affected customers had tenure = 0 months, indicating newly joined customers with no accumulated charges.
- The 11 blank TotalCharges values were replaced with 0, and the column was converted from object to float64.
- No fully duplicated rows were found.

2.2 Key EDA Findings

Feature / Segment	Churn Rate / Statistic	Interpretation
Month-to-month contract	42.7% churn	Much higher than one-year (11.3%) and two-year (2.8%) contracts
Fiber optic	41.9% churn	Higher than DSL (19.0%) and no internet (7.4%)
Electronic check	45.3% churn	Highest churn among payment methods
No technical support	41.6% churn	Compared with 15.2% for customers with technical support
No online security	41.8% churn	Compared with 14.6% for customers with online security
Tenure	Median: 10 vs 38 months	Churned customers tended to be newer
Monthly Charges	Median: 79.65 vs 64.43	Churned customers generally paid more each month

These are associations, not proof of causation. The model later evaluates how multiple variables work together for prediction.

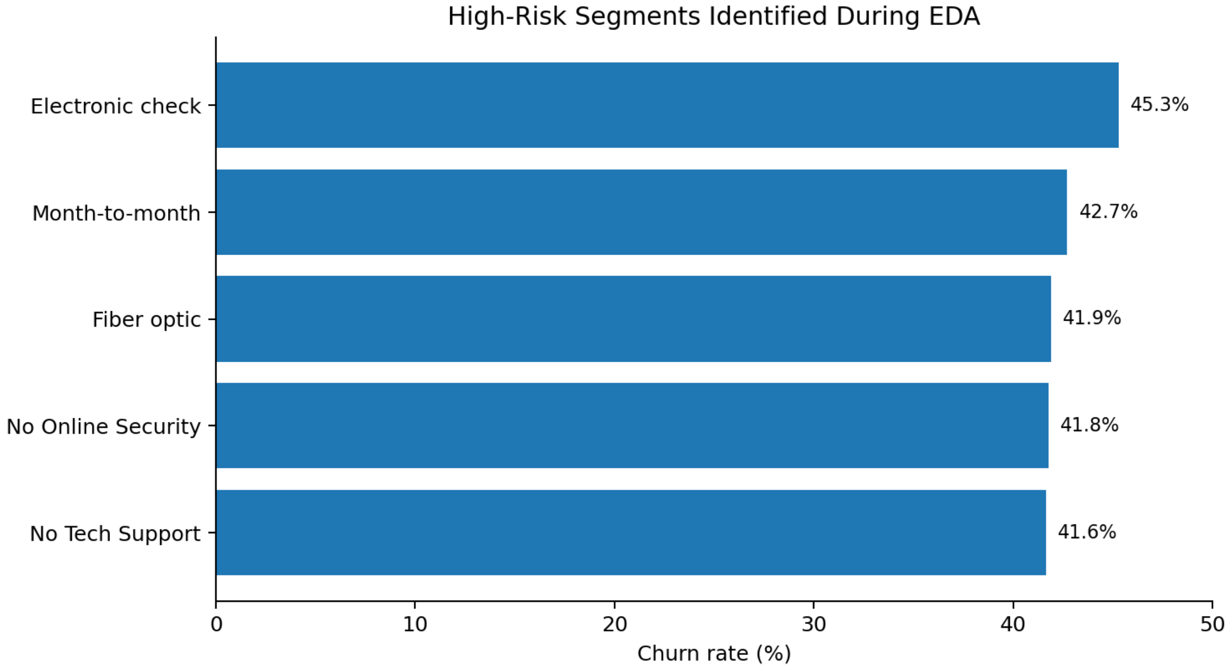

Figure 2. Selected high-risk segments identified during exploratory analysis.

3. Preprocessing and Modeling Strategy

3.1 Feature and Target Preparation

1. Remove customerID because it is an identifier rather than a meaningful predictive feature.
2. Separate input features X from the target y.
3. Encode the target as No = 0 and Yes = 1.
4. Create an 80/20 stratified train-test split so class proportions remain similar in both sets.
5. Identify categorical and numerical columns.
6. One-hot encode categorical variables while passing numerical columns through unchanged.

3.2 Pipeline and Leakage Prevention

A scikit-learn Pipeline combines the ColumnTransformer preprocessing step with each classifier. The encoder is fitted only on training data, then the learned transformation is applied to validation and test data. This reduces the risk of inconsistent preprocessing and data leakage.

Fit vs Transform

Fit learns preprocessing rules from training data. Transform applies those learned rules. Test data is transformed using training-fitted preprocessing and is not used to fit the encoder.

3.3 Validation Strategy

Model development used 5-fold cross-validation inside the training set. The final 20% test set was kept separate for holdout evaluation after model selection and hyperparameter tuning.

The majority-class DummyClassifier produced approximately 73.46% cross-validation accuracy, demonstrating why accuracy alone is not sufficient for this moderately imbalanced problem.

4. Model Development and Comparison

4.1 Models

- Decision Tree - a single interpretable tree that recursively splits data.
- Random Forest - an ensemble of diverse decision trees whose predictions are aggregated.
- XGBoost - a boosted tree ensemble built sequentially, where later trees improve errors left by earlier trees.

4.2 Cross-Validation Results

Model	Accuracy	Precision	Recall	F1
Decision Tree	72.15%	47.63%	48.83%	48.19%
Random Forest	78.70%	63.07%	47.76%	54.33%
XGBoost	78.42%	60.96%	51.97%	56.09%

Random Forest achieved the highest accuracy and precision. XGBoost achieved the highest recall and F1-score, which made it the preferred candidate for further tuning because identifying true churners is valuable for retention decisions.

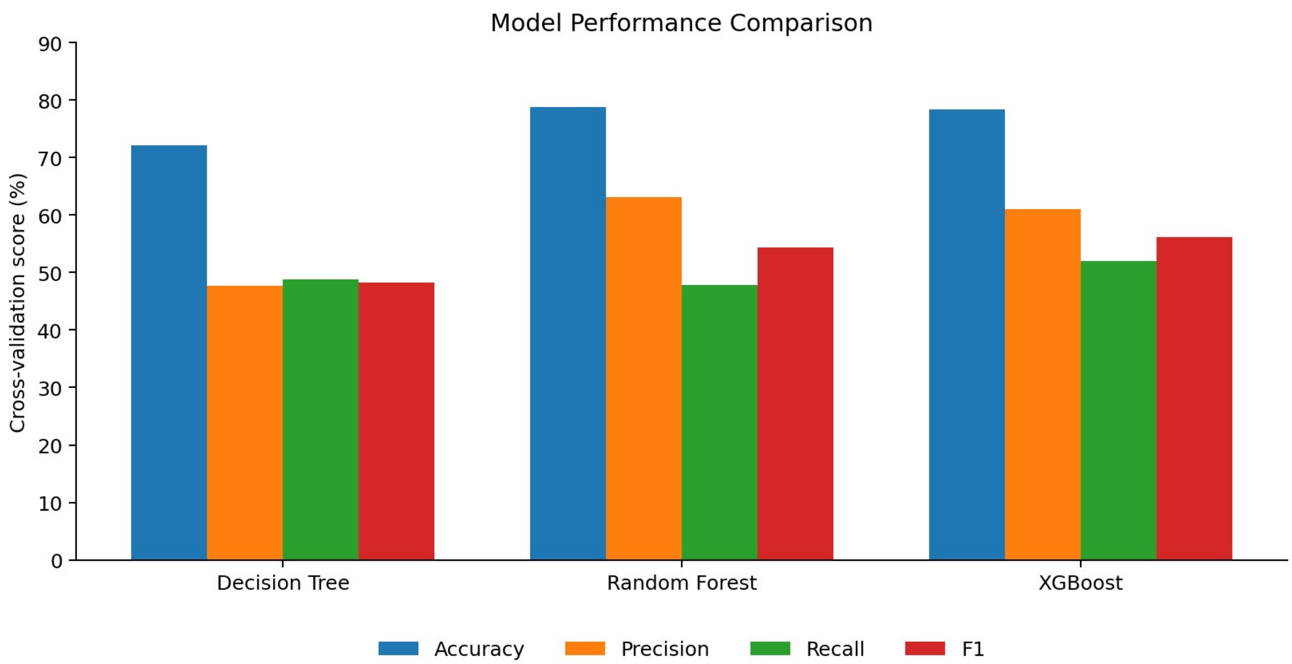

Figure 3. 5-fold cross-validation model comparison.

5. XGBoost Hyperparameter Tuning

RandomizedSearchCV was used to evaluate 20 random hyperparameter combinations with 5-fold cross-validation. F1-score was used as the optimization metric to balance precision and recall.

Hyperparameter	Selected Value	Role
n_estimators	300	Number of boosting trees
max_depth	2	Maximum complexity of each tree
learning_rate	0.05	Contribution of each new tree
subsample	0.8	Fraction of training rows used per tree
colsample_bytree	0.7	Fraction of features considered per tree

The selected configuration uses many shallow trees with a small learning rate. This is a controlled boosting strategy that can capture useful patterns while limiting the complexity of individual trees.

Tuning Improvement

Untuned XGBoost mean CV F1: 0.561. Best tuned mean CV F1: 0.595. Improvement: approximately 3.4 percentage points.

6. Final Evaluation on the Holdout Test Set

After tuning was completed using only the training data, the winning XGBoost pipeline was evaluated once on the untouched 20% test set.

Metric	Final Test Result
Accuracy	79.91%
Precision	65.85%
Recall	50.53%
F1-score	57.19%

The test F1-score was reasonably close to the tuned cross-validation F1-score, indicating that model performance transferred reasonably well to unseen customers. The model was correct about two-thirds of the time when it predicted churn, but it identified only about half of all actual churners.

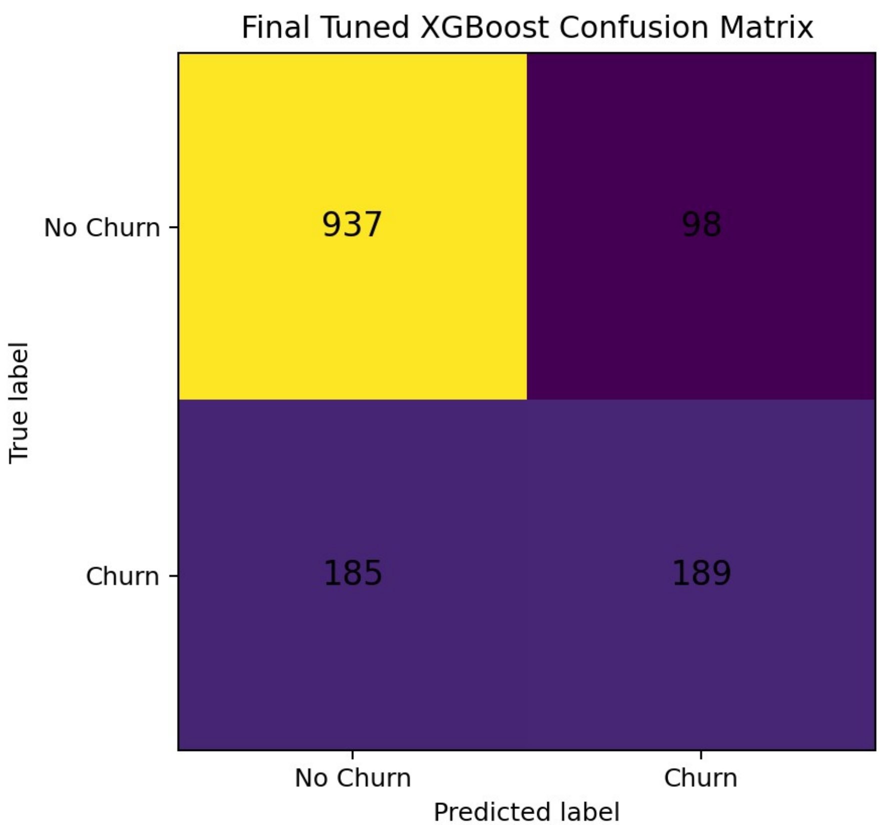

Figure 4. Final confusion matrix: TN=937, FP=98, FN=185, TP=189.

6.1 Business Meaning of the Confusion Matrix

- True Positive (189): correctly identified a customer who churned.
- False Positive (98): predicted churn for a customer who stayed; may lead to an unnecessary retention offer.
- False Negative (185): predicted no churn for a customer who actually left; this is a missed retention opportunity.
- True Negative (937): correctly identified a customer who stayed.

7. Feature Importance and Business Insights

Feature importance was extracted from the fitted tuned XGBoost model after retrieving the transformed feature names from the preprocessing pipeline.

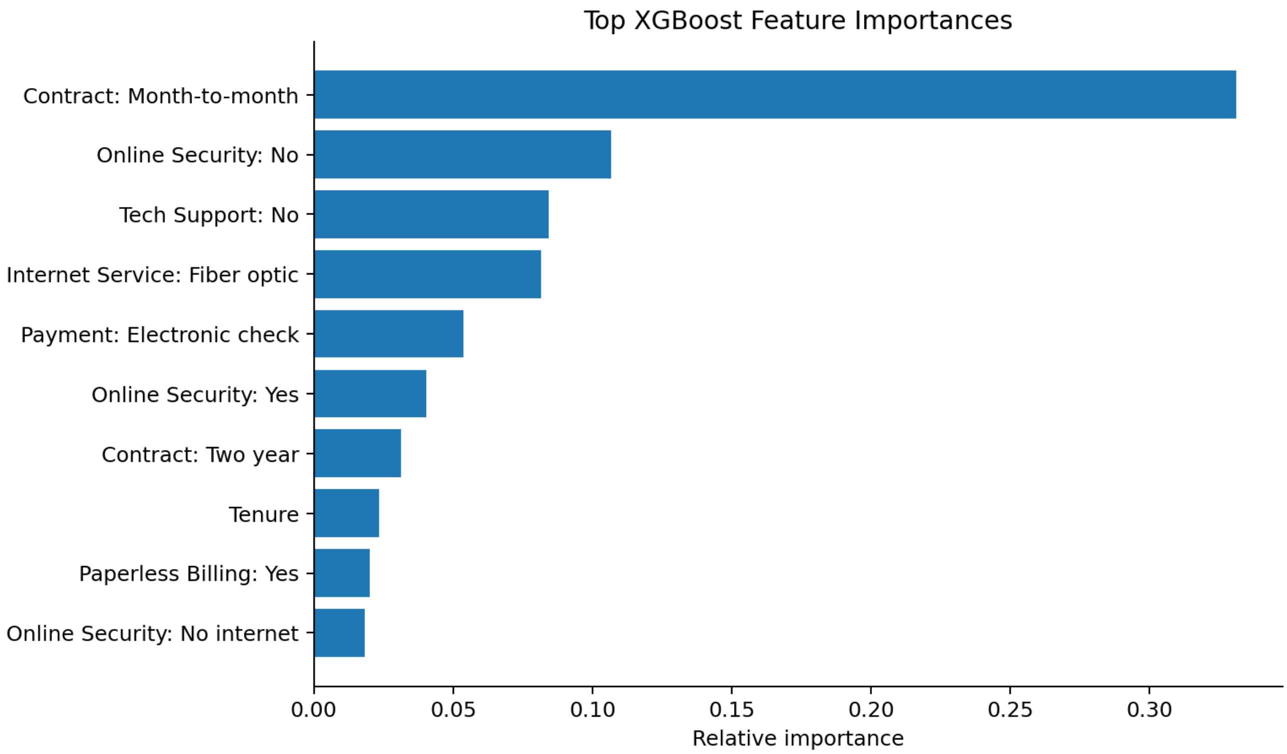

Figure 5. Top feature importances from the tuned XGBoost model.

7.1 Main Insights

- Month-to-month contracts were the most influential predictor by a large margin.
- Absence of online security and technical support were also important model signals.
- Fiber-optic internet and electronic-check payment were among the strongest categorical predictors.
- Tenure and total charges contributed to prediction, but their individual importance scores were lower than the strongest categorical indicators.

Feature importance indicates predictive usefulness, not causality. The findings are consistent with the EDA, which increases confidence that the model is learning patterns that are interpretable in the context of the dataset.

7.2 Suggested Retention Focus

A retention team could prioritize customers with combinations of high-risk characteristics, particularly newer month-to-month customers who use fiber-optic service, pay by electronic check, and lack technical support or online security. Any intervention strategy should be validated with controlled business experiments before treating these relationships as causal.

8. Limitations and Future Work

- Recall remains near 50%, so many real churners are still missed.
- The classification threshold remained at the default 0.5; lowering it could improve recall at the cost of more false positives.
- The model was evaluated on one historical dataset and has not been validated on future production data.
- Feature importance does not explain direction or causality; SHAP could provide more detailed local and global explanations.
- Business cost was not explicitly modeled. A cost-sensitive threshold should consider the cost of a false positive versus the cost of losing a churner.
- Additional feature engineering and alternative class-imbalance strategies could be explored.

9. Conclusion

This project demonstrates a complete and reproducible machine learning workflow for telecom customer churn prediction. It included data quality validation, EDA, preprocessing with a leakage-safe pipeline, a stratified holdout split, 5-fold cross-validation, comparison of multiple tree-based classifiers, hyperparameter tuning, final holdout evaluation, confusion-matrix analysis, and feature importance interpretation.

The tuned XGBoost model achieved 79.9% accuracy, 65.9% precision, 50.5% recall, and a 57.2% F1-score on the unseen test set. These results show meaningful predictive value, particularly when compared with the majority-class baseline, but the recall also shows that the model is not yet sufficient for an automated high-stakes retention workflow without additional optimization and business validation.

The most consistent churn signals across both EDA and model interpretation were month-to-month contracts, electronic-check payments, fiber-optic internet, lack of technical support, lack of online security, shorter tenure, and higher monthly charges. These findings can guide targeted retention analysis while remaining careful not to interpret association as causation.

Final Project Statement
The project is portfolio-ready as an end-to-end machine learning case study. Before production deployment, the next priorities would be threshold optimization, business-cost evaluation, stronger explainability, and validation on new data.